Supplementary Materials

Seizure Forecasting and Crash Risk Analysis

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1 Overview

This appendix states the model, derives its parameters, reports the main result and three sensitivity analyses, and gives reproduction instructions. All calculations are closed-form; the accompanying Jupyter notebook `crashes_vs_TiW.ipynb` reproduces every figure and number below.

We quantify the trade-off between seizure-forecasting accuracy (AUC), driving exposure, and the driving restrictions needed to hold fatal crash risk at or below a safety threshold of about 6× baseline. That threshold corresponds to a blood alcohol concentration (BAC) of roughly 0.05–0.07% for adult drivers, aligning with the 0.05% standard used in Utah and most European countries and is more conservative than the 0.08% limit in 49 U.S. states.

Relative fatal crash risk on forecast-safe days is computed from a per-hour fatal-crash baseline $\lambda_f \approx 4.5 \times 10^{-7}\,\mathrm{hr}^{-1}$ and a conditional probability $q_f = P(\text{fatal crash} \mid \text{seizure while driving}) = 0.02$ (central estimate; range 0.005–0.05). The chance that a seizure coincides with driving scales as $D/24$ for daily driving duration D, but because seizure-related and baseline risk both scale with exposure, the relative-risk benchmark is nearly invariant to D (Section 2.4).

Patients with infrequent seizures (≤ 1/year) approach unrestricted driving even at low AUC, while patients with frequent seizures need AUC ≥ 0.90 to drive meaningfully. At AUC 0.90 and 1 hr/day of driving, weekly seizures permit about 53 days/year, monthly seizures about 183, and yearly seizures about 365. The results depend little on driving duration, moderately on q_f, and substantially on the shape of the ROC curve at fixed AUC (Section 4).

2 Model and Methods

2.1 Forecast Scores and ROC Geometry

We model the forecasting algorithm's output as a scalar score S drawn from one of two normal distributions according to whether the day is a seizure day:

$$S \sim N(0,1) \quad \text{on a non-seizure day,} \qquad S \sim N(d',1) \quad \text{on a seizure day.} \tag{2.1}$$

The separation d' is the standard discriminability index of signal detection theory. The algorithm warns *unsafe to drive* when $S > t$ and forecasts *safe to drive* when $S \leq t$, for an operating threshold t. Writing Φ for the standard normal CDF and Φ^{-1} for its inverse, d' relates to AUC by

$$d' = \sqrt{2}\,\Phi^{-1}(\mathrm{AUC}), \tag{2.2}$$

so AUC 0.90 corresponds to $d' \approx 1.81$.

Two conditional probabilities carry the entire analysis: the *false-negative rate* $\mathrm{FNR}(t) = P(S \leq t \mid \text{seizure})$, the chance the algorithm calls a seizure day safe, and the *true-negative rate* $\mathrm{TNR}(t) = P(S \leq t \mid \text{no seizure})$, the chance it correctly calls a non-seizure day safe. Under (2.1) these are

$$\mathrm{FNR}(t) = \Phi(t - d'), \qquad \mathrm{TNR}(t) = \Phi(t). \tag{2.3}$$

Sensitivity is $1 - \mathrm{FNR}$ and the false-positive rate is $1 - \mathrm{TNR}$; neither is needed again. Figure S1 shows the geometry. Section 4.3 tests robustness to alternative ROC shapes, which replace (2.3) while leaving everything downstream unchanged.

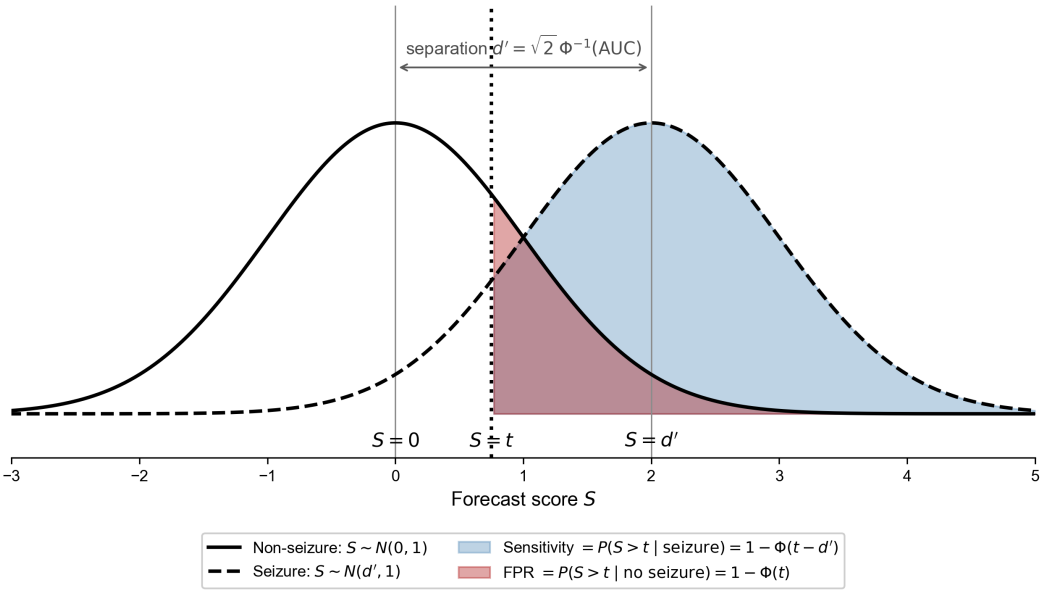


Figure S1: **Binormal forecast model.** Score distributions on non-seizure days ($S \sim N(0,1)$, solid) and seizure days ($S \sim N(d',1)$, dashed). The threshold t (dotted vertical) defines the decision boundary. Shaded areas right of t are sensitivity (blue) and the false-positive rate (red); the complementary areas left of t are FNR and TNR. The separation d' relates to AUC via Equation (2.2).

2.2 Daily Seizure Probability

For a patient with average seizure rate R (seizures per day), the daily seizure probability under a Poisson process is

$$p_{\mathrm{d}} = 1 - e^{-R}. \tag{2.4}$$

Frequency	R	p_{d}
1 seizure / week	1/7	0.1331
1 seizure / month	1/30	0.0328
1 seizure / year	1/365	0.0027

This mapping translates an average rate into a daily probability; the analysis is otherwise agnostic to the temporal structure of seizure risk.

2.3 Residual Seizure Probability on Forecast-Safe Days

When a patient drives on a forecast-safe day, the residual crash risk is governed by p_{res}, the probability that a seizure occurs despite the safe forecast. By Bayes' theorem,

$$p_{\mathrm{res}} \;=\; P(\text{seizure} \mid \text{forecast safe}) \;=\; \frac{\mathrm{FNR}\, p_{\mathrm{d}}}{\mathrm{FNR}\, p_{\mathrm{d}} + \mathrm{TNR}\,(1 - p_{\mathrm{d}})}. \tag{2.5}$$

The denominator is itself the probability that any given day is forecast safe,

$$P(\text{forecast safe}) \;=\; \mathrm{FNR}\, p_{\mathrm{d}} + \mathrm{TNR}\,(1 - p_{\mathrm{d}}), \tag{2.6}$$

so driving days per year are $365 \times P(\text{forecast safe})$ and warning days per year are the complement. Substituting (2.3) gives the closed forms used in the code; every later expression depends on the score model only through (2.5) and (2.6).

2.4 Fatal Crash Risk and Relative Risk

Table 1: Crash-risk model parameters.

Symbol	Meaning	Value	Source
λ_f	Baseline fatal-crash rate per hour of sober driving	$4.5 \times 10^{-7}\,\text{hr}^{-1}$	NHTSA *Passenger Vehicles 2023* + FHWA *Highway Statistics 2023* + AAA *American Driving Survey 2023* (derived below)
q_f	P(fatal crash | seizure while driving)	0.02 central; range 0.005–0.05	Inferred severity parameter; Section 4.1
D	Driving hours per day	0.5, 1, 2	—
p_{res}	P(seizure | forecast safe)	Equation (2.5)	—

Baseline per-day risk. Given D hours of driving per day, a sober driver faces a per-day baseline fatal-crash risk of

$$B(D) \;=\; D \cdot \lambda_f. \tag{2.7}$$

Derivation of λ_f. NHTSA *Passenger Vehicles 2023* reports about 44,000 passenger vehicles involved in fatal crashes; FHWA *Highway Statistics 2023* Table VM-1 reports about 2.88 trillion light-duty vehicle-miles traveled. The ratio is about 1.5 passenger-vehicle fatal-crash involvements per 10^8 miles. Converting per-mile to per-hour requires a driving speed: the AAA *American Driving Survey 2023* reports about 29.1 miles over about 60.7 minutes per driver per day, i.e. about 29 mph. Then

$$\lambda_f \;\approx\; 1.5 \times 10^{-8}\,\text{mile}^{-1} \times 29\,\text{mph} \;\approx\; 4.5 \times 10^{-7}\,\text{hr}^{-1}. \tag{2.8}$$

Because this baseline already includes all non-seizure causes of fatal crashes, the model conservatively charges it to every driving day regardless of seizure status.

Seizure-day risk. On a day when a seizure occurs, assuming uniform seizure timing across the 24-hour day,

$$P(\text{fatal crash} \mid \text{seizure day}) \;=\; \frac{D}{24} \cdot q_f \;+\; \left(1 - \frac{D}{24}\right) \cdot B(D), \tag{2.9}$$

where the first term covers seizures during driving and the second the baseline risk when the seizure falls outside driving hours.

Forecast-safe-day risk. On a forecast-safe day, residual fatal-crash risk is

$$P(\text{fatal crash} \mid \text{forecast safe}) \;=\; p_{\text{res}} \cdot P(\text{fatal crash} \mid \text{seizure day}) \;+\; (1 - p_{\text{res}}) \cdot B(D). \tag{2.10}$$

Relative risk and exposure invariance. Substituting (2.9) into (2.10) and dividing by $B(D)$ defines the relative risk:

$$\text{RR}(D) \;=\; \frac{P(\text{fatal crash} \mid \text{forecast safe})}{B(D)} \;=\; \frac{p_{\text{res}}\left[\dfrac{D}{24}\, q_f + \left(1 - \dfrac{D}{24}\right) B(D)\right] + (1 - p_{\text{res}})\, B(D)}{B(D)}. \tag{2.11}$$

Distributing the numerator and dividing each term by $B(D) = D\lambda_f$,

$$\text{RR}(D) \;=\; p_{\text{res}} \cdot \frac{(D/24)\, q_f}{D\, \lambda_f} \;+\; p_{\text{res}}\left(1 - \frac{D}{24}\right) \;+\; (1 - p_{\text{res}}). \tag{2.12}$$

The D in the first term cancels, and p_{res} cancels against $-p_{\text{res}}$, leaving the exact expression

$$\boxed{\text{RR}(D) \;=\; 1 + p_{\text{res}}\left[\, \frac{q_f}{24\, \lambda_f} - \frac{D}{24}\right]}. \tag{2.13}$$

Under our parameters $q_f/(24\lambda_f) \approx 1852$, while $D/24 \le 0.083$ for $D \le 2$ hours, so the $D/24$ term is negligible and

$$\text{RR}(D) \;\approx\; 1 + p_{\text{res}} \cdot A, \qquad A \;\equiv\; \frac{q_f}{24\, \lambda_f}, \tag{2.14}$$

where A is the amplification factor. Absolute fatal-crash risk still scales with driving exposure through $B(D)$, but relative risk against the BAC-anchored threshold does not. Section 4.2 confirms this numerically.

2.5 Safety Threshold and Operating Point

Under the doubling-per-0.02-BAC model of Zador et al. and Compton & Berning, the relative fatal-crash risk at BAC 0.05% is

$$T \;=\; 2^{\,0.05/0.02} \;\approx\; 5.66. \tag{2.15}$$

We refer to this as the "6× threshold" for readability; the code uses $T = 2^{0.05/0.02}$ exactly. Combining (2.13) with $\text{RR} \le T$ gives a ceiling on the residual seizure probability,

$$p_{\text{res}} \;\le\; p_{\text{res}}^{\max} \;=\; \frac{T - 1}{A - D/24}, \tag{2.16}$$

which equals 0.00251 at $q_f = 0.02$ and $D = 1\,\text{hr}$. Because p_{res} is increasing in t under the equal-variance model, the least restrictive admissible operating point is the largest t satisfying (2.16). We solve for it by root-finding rather than by scanning a grid of thresholds, which matters for the extreme operating points discussed next.

2.6 Feasibility of the Safety Constraint

Two boundary cases are easy to misread and are worth stating explicitly.

First, no AUC above 0.5 makes the constraint unattainable under the equal-variance model. As $t \to -\infty$, the ratio $\Phi(t - d')/\Phi(t) \to 0$, so $p_{\text{res}} \to 0$ and (2.16) is eventually satisfied for any $d' > 0$. What fails at low AUC is not safety but usefulness: at AUC 0.60 with weekly seizures the constraint is first met at $t \approx -11.2$, which permits on the order of 10^{-27} driving days per year. Such cells should be read as "permits effectively no driving," not as "cannot reach the safety threshold." Section 4.3 describes a case where infeasibility is genuine.

Second, no seizure frequency considered here is safe without any restriction at all. A patient who drives every day has $p_{\text{res}} = p_{\text{d}}$, so from (2.14) the unrestricted relative risk is $1 + p_{\text{d}}A$: about 248× baseline at one seizure per week, 62× at one per month, and 6.07× at one per year. Even the yearly figure exceeds $T = 5.66$. A patient with one seizure per year and an AUC 0.90 forecast therefore still requires about 0.33 warning days per year, and the entry "365 driving days" in the tables below is a rounded 364.7, not an exemption from restriction.

3 Main Result

Table 2 reports the driving restrictions required to meet the safety threshold, at 1 hr/day driving, $q_f = 0.02$, and $\lambda_f = 4.5 \times 10^{-7}\,\text{hr}^{-1}$.

Table 2: Driving restrictions required to meet the safety threshold, by seizure frequency and forecasting performance (1 hr/day driving). Driving days below 1 are shown in scientific notation: the safety threshold is met, but only by driving essentially never.

Seizure freq.	AUC	Warning days/yr	Driving days/yr	Avg. days between drives
1/week	0.60	365.0	6×10^{-27}	effectively never
	0.70	365.0	9×10^{-5}	effectively never
	0.80	363.5	1.5	236
	0.90	311.8	53.2	6.9
	0.95	219.7	145.3	2.5
	0.99	91.0	274.0	1.3
1/month	0.60	365.0	7×10^{-10}	effectively never
	0.70	364.3	0.7	495
	0.80	323.2	41.8	8.7
	0.90	181.8	183.2	2.0
	0.95	91.3	273.7	1.3
	0.99	23.0	342.0	1.1
1/year	0.60	33.4	331.6	1.10
	0.70	8.1	356.9	1.02
	0.80	2.0	363.0	1.01
	0.90	0.33	364.7	1.001
	0.95	0.12	364.9	1.000
	0.99	0.08	364.9	1.000

Figure S2 shows the mechanism: how crash risk on forecast-safe days falls as the algorithm spends more days in warning, and where each AUC curve crosses the safety threshold.

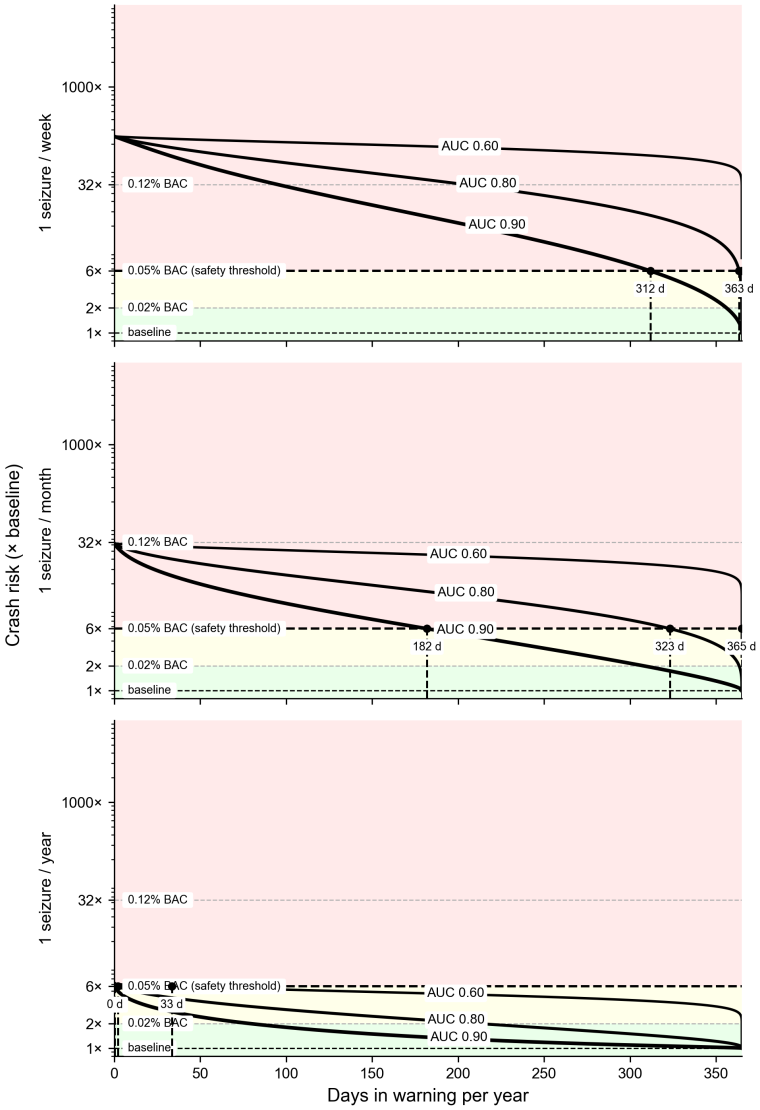


Figure S2: **Crash risk versus days in warning.** Rows are seizure frequencies (1/week, 1/month, 1/year), at 1 hr/day driving. Black curves are AUC 0.60, 0.80, 0.90 (thicker lines indicate higher AUC). Shaded bands and dashed lines mark BAC-equivalent risk levels (0.02%, the 0.05% safety threshold, 0.12%). Black circles mark the minimum days in warning needed to reach the threshold.

Four points follow. Algorithms need AUC ≥ 0.90 to provide meaningful driving days for patients with weekly or monthly seizures, and even that is necessary rather than sufficient. Patients with rare seizures (≤ 1/year) are near unrestricted driving even at AUC 0.60 (331.6 days/year), subject to the caveat in Section 2.6 that some restriction is always required. Patients with monthly seizures gain the most from better algorithms: AUC 0.90 permits about 183 days/year and AUC 0.95 about 274. Patients with weekly seizures face heavy restrictions throughout: AUC 0.90 permits only about 53 days/year, and AUC 0.80 is of no practical use.

4 Sensitivity Analyses

4.1 Conditional Fatal-Crash Probability q_f

The parameter $q_f = P(\text{fatal crash} \mid \text{seizure while driving})$ is the model's most uncertain component. No study reports it directly, so we treat it as an inferred severity parameter and examine the defensible range $[0.005,\ 0.05]$:

- Lower bound, $q_f = 0.005$: a moderate crash rate given a seizure while driving (~ 0.30) combined with moderate case-fatality (~ 0.02), closer to baseline fatal-crash severity.
- Central, $q_f = 0.02$: the Gastaut & Zifkin 1987 observed crash rate (~ 0.55) combined with case-fatality of seizure-related crashes ($\sim$ 3–5%, inferred from Sheth et al. 2004 and Drazkowski et al. 2003).
- Upper bound, $q_f = 0.05$: counting near-misses as effective crashes (Gastaut reports a further $\sim 38\%$ near-miss rate) with upper-bound severity.

Table 3: Sensitivity of driving days/year to q_f at AUC = 0.90, 1 hr/day.

Seizure frequency	$q_f = 0.005$	$q_f = 0.02$ (central)	$q_f = 0.05$
1/week	154.8	53.2	19.8
1/month	314.3	183.2	99.1
1/year	365.0*	364.7	334.5

*The only cell in this analysis where the threshold is met with no restriction at all: at $q_f = 0.005$ the unrestricted relative risk for a yearly-seizure patient is 2.3× baseline, below T.

The amplification factor $A = q_f/(24\lambda_f)$ runs from 463 (lower) through 1852 (central) to 4630 (upper), so the ceiling $p_{\text{res}}^{\max}$ falls from 1.01% to 0.10%. Across this range the qualitative finding holds: weekly-seizure patients cannot achieve unrestricted driving even at AUC 0.90, and yearly-seizure patients stay near unrestricted driving. The allowance for monthly seizures spans about a factor of three; a policy that errs toward caution should use $q_f = 0.05$ until empirical data anchor the parameter better.

4.2 Driving Duration

Section 2.4 shows analytically that relative risk is nearly invariant to driving duration D, because seizure-related and baseline fatal risk both scale with exposure. Table 4 confirms this numerically. At the reported precision of whole days, the results are identical across 0.5, 1, and 2 hours of driving per day; the largest difference anywhere in the table is 0.003 days per year, about five minutes.

Absolute fatal-crash risk does scale with driving duration, and an *absolute*-risk benchmark would depend directly on exposure. Under the relative-risk benchmark used here, driving duration is not a material determinant of allowable driving days.

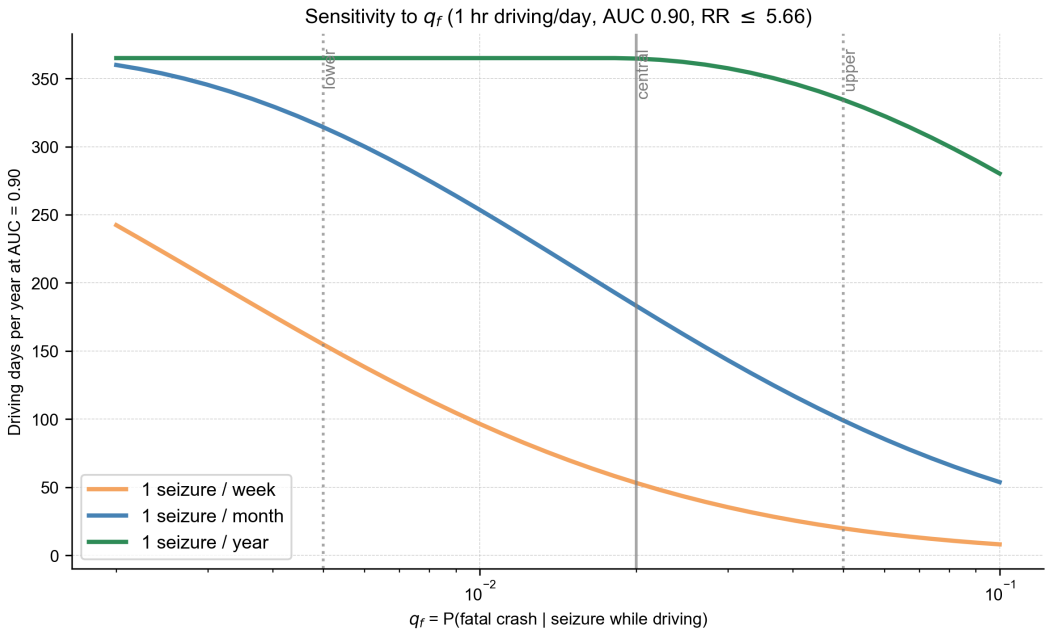


Figure S3: **Sensitivity of driving days to q_f.** Driving days per year at AUC = 0.90 and 1 hr/day, as a function of q_f. Vertical guides mark the lower bound (0.005), the central estimate (0.02), and the upper bound (0.05).

Table 4: Driving days/year across exposure levels at AUC = 0.90, shown to three decimals to make the residual D-dependence visible.

Seizure frequency	30 min/day	1 hr/day	2 hr/day
1/week	53.174	53.175	53.176
1/month	183.163	183.164	183.167
1/year	364.674	364.674	364.674

4.3 ROC Curve Shape

Real ROC curves deviate from the equal-variance form because of unequal variances (S-shaped) or mixture distributions (hooked, when some seizures are easier to predict than others). We compare three score-generating models, all calibrated to AUC = 0.90 exactly so that the comparison isolates shape. Table 5 lists their parameters and Figure S4 shows the resulting curves.

Only the FNR column of Table 5 changes between models; $\mathrm{TNR}(t) = \Phi(t)$ throughout, and Equations (2.5) and (2.6) carry through unchanged. Table 6 reports the resulting driving days, and Figure S5 the corresponding risk curves.

Why the S-shaped model can fail outright. The "not achievable" entry above is qualitatively different from the near-zero entries in Table 2. When $\sigma_1 > 1$, the seizure-day score distribution has the heavier lower tail, so as $t \to -\infty$ the ratio $\Phi\big((t-\mu_1)/\sigma_1\big)/\Phi(t) \to \infty$ and therefore $p_{\mathrm{res}} \to 1$. Tightening the threshold past a point makes forecast-safe days *more* seizure-prone rather than less, and $p_{\mathrm{res}}(t)$ attains an interior minimum, here at $t \approx -1.94$. That minimum is a hard floor on

Table 5: Parameters of the three ROC models. Non-seizure scores are $N(0,1)$ in all three. Each parameter set is calibrated to AUC $= 0.90$ exactly.

Model	Seizure-day score distribution	FNR$(t) = P(S \le t \mid \text{seizure})$
Equal-variance binormal	$N(d', 1)$, $d' = 1.812$	$\Phi(t - d')$
Unequal-variance (S-shaped)	$N(\mu_1, \sigma_1^2)$, $\mu_1 = 2.205$, $\sigma_1 = 1.4$	$\Phi\big((t - \mu_1)/\sigma_1\big)$
Mixture (hooked)	$w\,N(\mu_{\text{easy}}, 1) + (1-w)\,N(\mu_{\text{hard}}, 1)$, $w = 0.7$, $\mu_{\text{easy}} = 2.716$, $\mu_{\text{hard}} = 0.869$	$w\,\Phi(t - \mu_{\text{easy}}) + (1 - w)\,\Phi(t - \mu_{\text{hard}})$

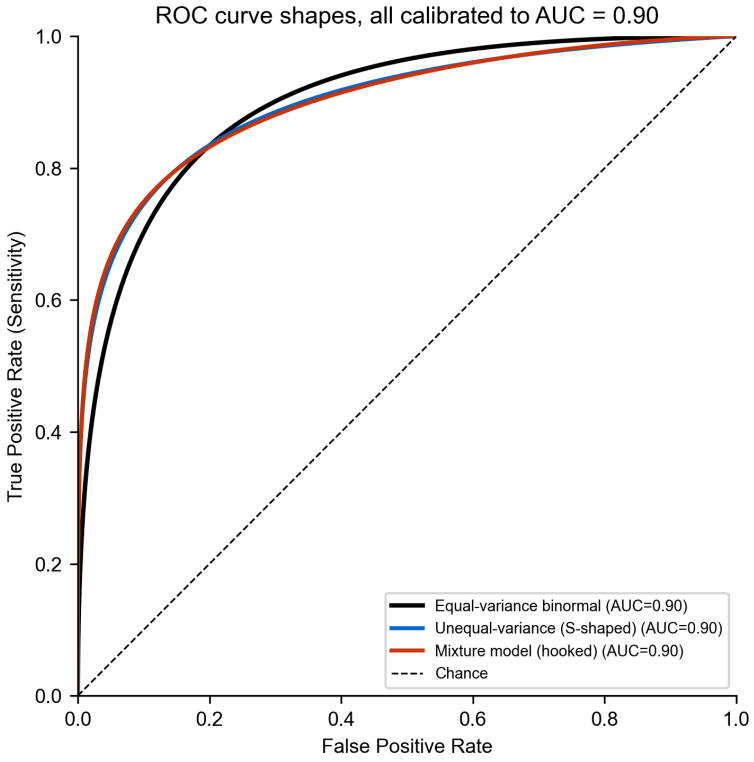


Figure S4: **Three ROC curve shapes**, all at AUC $= 0.90$. Equal-variance binormal (black), unequal-variance/S-shaped (blue), and hooked mixture (red).

Table 6: Driving days/year at AUC $= 0.90$, 1 hr/day, by ROC shape.

Seizure frequency	Equal-variance binormal	S-shaped (unequal var.)	Hooked (mixture)
1/week	53.2	not achievable	1.4
1/month	183.2	72.4	88.1
1/year	364.7	364.9	364.9

achievable residual risk: 0.0089 for weekly seizures, 0.0020 for monthly, 0.00016 for yearly, against a ceiling $p_{\text{res}}^{\max} = 0.0025$. Weekly seizures exceed the floor, so no operating point whatsoever satisfies the safety constraint. Restriction alone cannot fix a forecaster whose misses are concentrated in its most confident predictions.

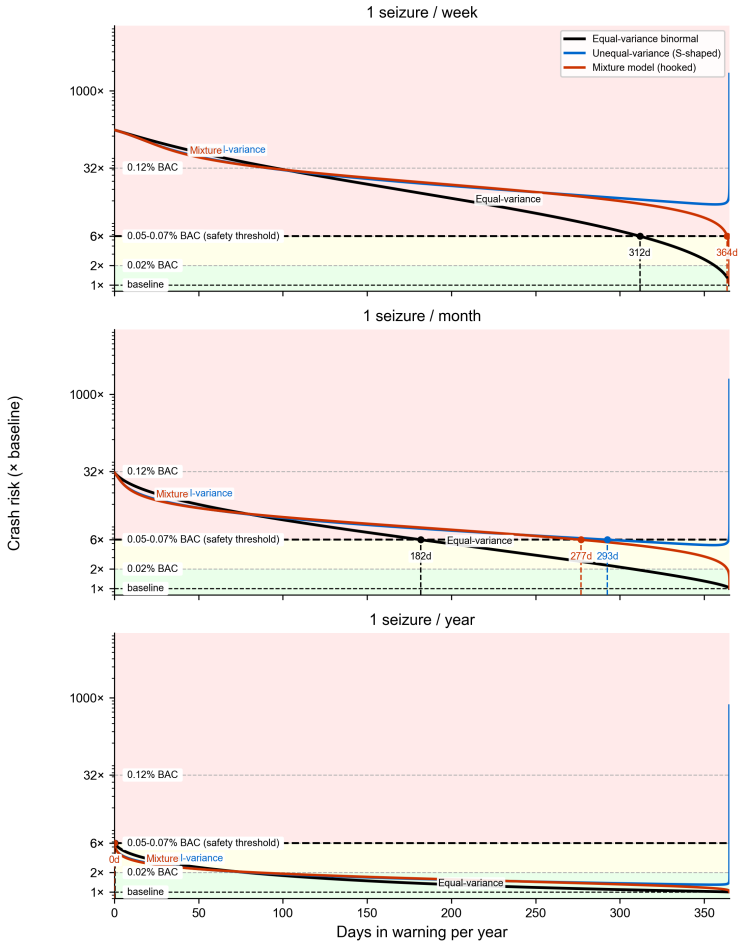


Figure S5: **Crash risk versus days in warning for three ROC shapes**, by seizure frequency, at 1 hr/day and AUC = 0.90. Equal-variance binormal (black), S-shaped (blue), and hooked mixture (red). Black dashed line: safety threshold.

At high seizure frequencies, shape matters as much as AUC. The equal-variance binormal is the most optimistic of the three; at one seizure per month the allowance falls from 183 days/year to 88 (hooked) or 72 (S-shaped), and at one per week the S-shaped model admits no safe operating point at all. At one seizure per year all three shapes permit essentially full-year driving, because the prior seizure probability is already low enough that the posterior clears the threshold under any of them. Since equal-variance binormal is the most optimistic case, real-world ROCs at the same AUC will generally permit fewer driving days than Table 2 suggests. Reporting AUC alone therefore overstates clinical utility for patients with frequent seizures, while the qualitative conclusion, that high AUC is necessary and that even AUC 0.90 is borderline for weekly seizures, holds under all three shapes.

5 Assumptions and Limitations

Why the binormal model is a reasonable base case. It is standard in diagnostic testing and signal detection theory, it is tractable in closed form, and for a given AUC it offers near-best-case sensitivity–specificity trade-offs, which gives the forecasting algorithm the benefit of the doubt. The question here is what level of performance is needed for safe driving decisions, not how one specific empirical ROC curve performs. Section 4.3 quantifies what happens when the assumption fails: departures reduce the allowable driving days rather than increasing them, so the assumption is optimistic and the reported allowances are upper bounds.

Driving-exposure assumptions.

1. Seizure timing is uniform over the 24-hour day. Many patients have circadian patterns such as nocturnal epilepsy or morning clustering. The model is conservative for seizures concentrated outside driving hours and optimistic for daytime clustering.
2. Driving duration is constant across days. Real driving varies; the 30 min/day to 2 hr/day range brackets typical commuter and heavy-driver patterns, and per Section 4.2 the choice barely affects the relative-risk result.
3. $q_f = 0.02$ is an inferred severity parameter (range 0.005–0.05), decomposing as $P(\text{crash} \mid \text{seizure while driving}) \approx 0.55$ (Gastaut & Zifkin 1987) times case-fatality $\approx$ 3–5% (inferred from Sheth et al. 2004 and Drazkowski et al. 2003). It depends on seizure type, aura, and road and vehicle conditions; Section 4.1 reports the sensitivity analysis.
4. A single fixed operating threshold is used year-round. Adaptive thresholds, or policies that condition on recent seizure history, are outside the scope of this model.

6 Recommendations

6.1 For algorithm developers

1. Treat AUC ≥ 0.90 as necessary but not sufficient for driving applications. Table 6 shows two forecasters at identical AUC 0.90 whose safe allowances differ by a factor of two, and one that admits no safe operating point at all.
2. Report the empirical ROC shape, not only its area. At minimum, report sensitivity and specificity at the operating points a driving policy would actually use, which are in the high-specificity region far from the curve's midpoint.
3. Check whether the model's misses concentrate in its confident predictions. That pattern produces the S-shaped failure of Section 4.3, in which no amount of added restriction reaches the safety threshold.
4. For patients with frequent seizures, expect AUC 0.90 to be insufficient for a practical driving policy, and calibrate development targets accordingly.

6.2 Clinical and policy implications

1. Counsel patients using seizure frequency and driving pattern together, since the same algorithm yields very different allowances across frequencies (Table 2).
2. For patients with rare seizures, forecasting adds little to driving safety, and its value should be judged on other quality-of-life grounds.
3. Do not read a rounded "365 driving days" as a finding of no restriction. Every seizure frequency considered here requires some restriction to meet the threshold (Section 2.6).
4. Any policy built on these numbers inherits the uncertainty in q_f, which spans a factor of three in allowance for monthly-seizure patients (Table 3).

7 Reproducibility

All calculations use closed-form expressions implemented with NumPy, SciPy (`scipy.stats.norm` for the Gaussian CDF and its inverse), and Matplotlib. Operating points are located with a bracketed root-find on (2.16) rather than by scanning a threshold grid. The distinction is not cosmetic: a grid both quantizes the reported allowance by about one day and truncates the extreme operating points, which is what produces spurious "cannot reach threshold" results at low AUC.

```
pip install numpy matplotlib scipy
jupyter notebook crashes_vs_TiW.ipynb        # interactive
# or run all cells programmatically:
jupyter nbconvert --to notebook --execute crashes_vs_TiW.ipynb
```

Table 7: Notebook cell layout and outputs.

Cell	Produces
0	Shared exact solver for the operating point
1	Figure S2 — crash risk vs. days in warning (3 frequencies, 1 hr/day)
2	Figure 1 — driving days vs. AUC (main result)
3	Figure S1 — binormal model schematic
4	Figures S4 & S5 — ROC shape geometries and sensitivity
5–6	Extended AUC table (tab-delimited, all frequencies and durations)
7	Figure S3 — q_f sensitivity

Figure files are produced as PDF and PNG: `Figure_1`, `Figure_S1`–`Figure_S5`.